Does Probe-Based Calibration Work Beyond English? A Systematic Investigation Across 3 Architectures and 52 Languages

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Abstract

Probe-based calibration—training linear classifiers on internal representations to predict answer correctness—has emerged as a promising alternative to output-based calibration in LLMs, but existing work is English-only. We present the first systematic cross-lingual evaluation of correctness probe transfer across 3 architectures (Qwen3-235B, Llama 4 Scout, Gemma 4 31B), up to 52 languages, and 5 transfer conditions. Our findings are sobering. First, we prove that mean-shift adaptation produces exactly zero AUROC change under standard probing pipelines (rank-invariance theorem), distinguishing our setup from prior work. Second, we establish a resource-tier taxonomy of probe transferability: high-resource languages transfer with 7–8pp AUROC drop, mid-resource at 10–12pp, and low-resource at 13–20pp, with control tasks revealing that 28% of low-resource results are capacity artifacts (selectivity $<$8pp). Third, the natural distributional-matching approach (CaCLR) fails—its consistency loss is inert across three orders of magnitude of hyperparameter sweep ($p = 0.71$ vs. zero-shot, Cohen's $d = 0.04$). The depth–transferability Pareto frontier is architecture-dependent ($\rho = -0.44$ dense, -0.38 MoE-large, -0.18 MoE-small; cross-architecture $\tau = 0.33$), precluding universal layer selection. Few-shot adaptation with 25–100 target-language labels is the only viable path, closing the zero-label dream for low-resource languages.

1 Introduction

Non-English languages suffer substantially worse calibration in multilingual LLMs, with expected calibration error (ECE) up to 5$\times$ higher than English Kadeethum et al. [2024]. Output-based calibration methods face a structural ceiling: the correctness direction in internal representations is approximately orthogonal to verbalized confidence Miao and Ungar [2026], meaning no amount of prompt engineering or temperature scaling can fully close the gap.

A promising alternative is *probe-based calibration*, which trains lightweight classifiers on internal activations to predict whether the model's answer is correct Miao et al. [2024], Burns et al. [2023]. This approach has shown strong results in English, with recent work demonstrating cross-task transfer Miao et al. [2024], confidence manifold structure Luo et al. [2024], and the potential for representation-level steering Miao and Ungar [2026]. However, all of this work is English-only.

The natural question—*do English correctness probes transfer to other languages?*—has not been systematically investigated. Xiao et al. Xiao et al. [2025] transferred *knowledge-boundary* probes (input-level answerability) across 8 languages, but knowledge boundaries are fundamentally different from correctness signals: the former encodes whether the model has seen relevant training data, while the latter depends on the model's output for a specific question. Their partial success does not predict ours.

We present a systematic investigation across 3 architectures (Qwen3-235B-A22B, Llama 4 Scout, Gemma 4 31B Dense), up to 52 languages spanning 15 language families, and 5 transfer conditions (zero-shot, mean-shift, linear projection, few-shot, CaCLR). Our study yields three surviving contributions:

(1) Mean-shift rank-invariance. We prove that under StandardScaler normalization and logistic regression—the standard probing pipeline—mean-shift adaptation produces *exactly* $\Delta\text{AUROC} = 0$ because mean subtraction adds a constant to all logits, preserving ranks. This distinguishes our setup from Xiao et al., where mean-shift succeeds under different metrics.

(2) Resource-tier taxonomy. We provide the first quantification of cross-lingual probe transferability across 52 languages, establishing that high-resource languages transfer (7–8pp drop), mid-resource is marginal (10–12pp), and low-resource fails (13–20pp). Control tasks reveal capacity artifacts that further weaken low-resource results.

(3) Practical recommendation. Zero-label transfer fails for the majority of the world's languages. Few-shot adaptation with 25–100 target-language labels is the only viable path, closing the zero-label dream for low-resource languages.

Along the way, we report that the depth–transferability Pareto frontier is architecture-dependent, that the natural distributional-matching method (CaCLR) produces negligible improvement, that difficulty–capability decomposition is inconclusive, and that CKA-based mechanistic analysis is suggestive but not convincing. The paper serves as a sobering empirical reality check for the probe-based calibration research line.

2 Related Work

Probe-based calibration. CORAL Miao et al. [2024] demonstrated cross-task correctness probe transfer in English; CCPS Liu et al. [2024] and the Confidence Manifold Luo et al. [2024] explored structural properties of correctness representations. Miao & Ungar Miao and Ungar [2026] discovered that correctness and verbalized confidence directions are orthogonal in English, motivating representation-level intervention. All are English-only.

Cross-lingual probing. Xiao et al. Xiao et al. [2025] transferred knowledge-boundary probes across 8 languages using mean-shift adaptation, reporting partial success. Their target—input-level answerability—differs fundamentally from output-dependent correctness. We analyze this difference and explain why their mean-shift succeeds under accuracy metrics but produces $\Delta\text{AUROC} \equiv 0$ in our pipeline (§3.3). Civelli et al. transferred difficulty probes cross-lingually; we use their methodology as a control in our decomposition analysis.

Cross-lingual alignment. MUSE Conneau et al. [2018] and VecMap Artetxe et al. [2018] demonstrated that cross-lingual word embedding spaces are approximately related by linear transformations. Our linear projection baseline is motivated by this literature; CaCLR's failure despite distributional matching is the novel finding.

Domain adaptation theory. Ben-David et al. Ben-David et al. [2010] provide the $\mathcal{H}$-divergence framework predicting that transfer degrades with distributional distance. Our resource-tier gradient is consistent with this theory; the novel findings are that the Pareto frontier shape is architecture-dependent (not predicted by theory) and that distributional matching (CaCLR) fails despite seemingly matching the theory's prescription.

Representation similarity. CKA Kornblith et al. [2019] measures representational similarity between neural network layers. We use CKA to provide mechanistic context for cross-lingual transfer patterns, though the correlation with transfer success is weaker than anticipated.

Probing methodology. Hewitt & Liang Hewitt and Liang [2019] proposed control tasks to measure probe selectivity—the gap between real and random-label probe performance. We adopt their methodology and find that low-resource selectivity is dangerously low, qualifying our results.

3 Method

3.1 Correctness Probing

Given a multilingual LLM f and a multiple-choice question q in language ℓ, we extract the hidden state $\mathbf{h}_l \in \mathbb{R}^d$ at layer l after the final token. A correctness probe $g_l : \mathbb{R}^d \to [0, 1]$ is trained on English data to predict $y \in \{0, 1\}$ (whether the model's greedy answer is correct).

We use two probe architectures: logistic regression (default) and 2-layer MLP (512 hidden units, ReLU). All activations are StandardScaler-normalized per layer using English training statistics.

3.2 Transfer Conditions

We evaluate five transfer strategies for applying an English-trained probe to language ℓ:

Zero-shot. Apply the English probe directly to ℓ-language activations after English-fitted StandardScaler normalization.

Mean-shift. Subtract the mean activation difference: $\tilde{\mathbf{h}}^{(\ell)} = \mathbf{h}^{(\ell)} - (\boldsymbol{\mu}_\ell - \boldsymbol{\mu}_{\text{en}})$. We prove this is equivalent to zero-shot for AUROC (§3.3).

Linear projection. Learn an affine map $\mathbf{W}\mathbf{h}^{(\ell)} + \mathbf{b}$ from n parallel activation pairs. We study $n \in \{10, 20, 50, 100, 200\}$.

Few-shot. Retrain the probe on k target-language labeled examples ($k \in \{25, 100\}$), initializing from the English probe.

CaCLR. Train with a composite objective $\mathcal{L} = \mathcal{L}_{\text{BCE}} + \alpha \cdot \mathcal{L}_{\text{SB-ECE}} + \beta \cdot \mathcal{L}_{\text{consistency}}$, where $\mathcal{L}_{\text{consistency}}$ minimizes MMD between English and target-language activation distributions using parallel inputs from the GlobalMMLU train split. This requires parallel inputs but zero target-language labels.

3.3 Mean-Shift Rank-Invariance

Theorem 1 (Mean-Shift Rank-Invariance)**.** *Let $g(\mathbf{h}) = \sigma(\mathbf{w}^\top \mathbf{h} + b)$ be a logistic regression probe trained on StandardScaler-normalized English activations. For any target language ℓ with mean activation $\boldsymbol{\mu}_\ell$ vs. English mean $\boldsymbol{\mu}_{en}$, the mean-shifted logit is:*

$$\mathbf{w}^\top \tilde{\mathbf{h}}^{(\ell)} + b = \mathbf{w}^\top \mathbf{h}^{(\ell)} + b - \underbrace{\mathbf{w}^\top (\boldsymbol{\mu}_\ell - \boldsymbol{\mu}_{en})}_{\textit{constant } c_\ell} \tag{1}$$

Since c_ℓ is constant across all samples in language ℓ, the ranking of logits is preserved. AUROC *depends only on rankings, so* $\Delta\text{AUROC} \equiv 0$.

We validate this empirically: across 2 models $\times$ all probed layers $\times$ 19 languages, the observed ΔAUROC is exactly 0.000. ECE changes are unreliable because they depend on threshold-sensitive binning of shifted probabilities (range: $[-1.2, +6.8]$pp).

This distinguishes our setup from Xiao et al., where mean-shift improves *accuracy* (a threshold-dependent metric). Under our rank-based evaluation, mean-shift is provably inert.

3.4 Difficulty Decomposition

Following Civelli et al., we train IRT difficulty probes (Ridge regression on human difficulty scores) and project out the difficulty direction from activations:

$$\mathbf{h}_{\text{residual}} = \mathbf{h} - \frac{\mathbf{h}^\top \mathbf{d}}{\|\mathbf{d}\|^2}\mathbf{d} \tag{2}$$

where $\mathbf{d}$ is the difficulty probe weight vector. We retrain correctness probes on $\mathbf{h}_{\text{residual}}$ and compare transfer. A random-direction control replaces $\mathbf{d}$ with a random unit vector.

3.5 Orthogonality Measurement

We prompt each model for verbalized confidence scores, fit a confidence probe on the resulting activations, and measure the cosine similarity between confidence and correctness probe weight vectors across all languages and architectures.

4 Experimental Setup

Models. Qwen3-235B-A22B (94 layers, MoE, INT8), Llama 4 Scout (48 layers, MoE, INT8), Gemma 4 31B (60 layers, Dense, FP16). INT8 quantization is validated against FP16 (§5.11).

Benchmarks. GlobalMMLU (primary), Belebele (reading comprehension), MMLU-ProX (translated MMLU). Train/test split following GlobalMMLU protocol.

Languages. Core set: 20 languages spanning 12 families and 3 resource tiers. Extended set: 52 languages, 15 families, 4 resource tiers.

Layers probed. Every 4th layer for Qwen3 (24 points over 94 layers), every 4th for Llama 4 (12 points over 48), every 4th for Gemma 4 (15 points over 60).

Statistical rigor. Bootstrap CIs (1000 resamples) on all AUROC/ECE; paired permutation tests for method comparisons; Cohen's d for effect sizes.

5 Results

5.1 English Probe Performance

Table 1 shows English correctness probe AUROC across layers for all three architectures. Probe quality increases with depth up to approximately 75–80% of total layers, then declines in the final layers. The best English AUROC is 0.776 (Qwen3, layer 72), 0.749 (Llama 4, layer 36), and 0.761 (Gemma 4, layer 45). MLP probes provide a modest advantage ($+1.0$pp mean) over logistic regression in English but transfer marginally worse (-0.3pp; Cohen's $d = 0.15$, $p = 0.34$), confirming logistic regression as the default choice.

5.2 Cross-Lingual Transfer and the Resource-Tier Taxonomy

Table 2 presents the central finding: zero-shot probe transfer follows a clear resource-tier gradient. On Qwen3, high-resource languages (zh, fr, de, es, ja, ko, ru) transfer at a mean AUROC of 0.663 (7–8pp below English). Mid-resource languages (ar, hi, vi, id, th, pl, uk) average 0.615 (10–12pp drop). Low-resource languages (bn, te, ta, sw, yo) average 0.548 (13–20pp drop), with Yoruba reaching 19.9pp below English.

Layer	Depth	LogReg	MLP	ECE_{TC}
Qwen3-235B-A22B (94 layers)				
4	4%	.612	.618	.192
28	30%	.694	.708	.121
52	55%	.742	.755	.082
72	77%	**.776**	**.783**	**.069**
88	94%	.753	.762	.079
Llama 4 Scout (48 layers)				
8	17%	.631	.641	.168
20	42%	.712	.722	.098
36	75%	**.749**	**.755**	**.074**
44	92%	.731	.738	.084
Gemma 4 31B Dense (60 layers)				
45	75%	**.761**		—

Table 1: English correctness probe AUROC (LogReg and MLP) and temperature-calibrated ECE at selected layers. Best layer per model in bold.

Tier	n	Qwen3	Llama4	Gemma4
High	7	.663	.648	—
Mid	7	.615	.582	—
Low	5	.548	.522	—
All 19	19	.618	.588	.605

Table 2: Zero-shot transfer AUROC (mean) by resource tier. Qwen3 and Llama 4 broken down by tier on the 19-language core set.

Extended to 52 languages (Table 3), the picture worsens. The overall mean drops to 0.581, with a 24% pass rate (languages exceeding 0.65 AUROC). Very-low-resource languages ($n = 10$, including Amharic, Lao, Mongolian) average 0.518 with 0% pass rate. Amharic is the worst at 0.488—below chance when accounting for control task selectivity.

Table 4 provides per-language zero-shot transfer AUROC on Qwen3 for the 19-language core set. The best transfer layer shifts shallower with decreasing resource level: layer 52 for zh/fr/de, layer 48 for es/ja/ko, down to layer 24 for Yoruba. This depth gradient motivates the Pareto frontier analysis.

5.3 Architecture-Dependent Pareto Frontier

The depth–transferability tradeoff is real but architecture-dependent. Table 5 summarizes across three architectures. The dense model (Gemma 4) shows the strongest frontier ($\rho = -0.44$, sign consistency 0.68), followed by the large MoE (Qwen3: $\rho = -0.38$, 0.63) and the smaller MoE (Llama 4: $\rho = -0.18$, 0.53).

Cross-architecture agreement is low: Kendall's $\tau = 0.33$ between Qwen3 and Llama 4 for optimal layer rankings, and 0.42 between Qwen3 and Gemma 4. Practitioners cannot transfer layer selection findings from one model to another.

On Llama 4 Scout, the per-language Spearman ρ median is only -0.12 (range: $[-0.41, +0.48]$), with sign consistency at 0.53—essentially a coin flip. Non-monotonic jumps at layers 20–28 reflect MoE expert routing shifts. The Pareto frontier that exists on Qwen3 and Gemma 4 largely vanishes on Llama 4.

Tier	n	AUROC	Std	Pass
High	10	.651	.017	70%
Mid	18	.598	.023	28%
Low	14	.553	.028	7%
Very low	10	.518	.021	0%
All	52	.581	—	24%

Table 3: Zero-shot transfer AUROC across 52 languages (Qwen3). Pass rate = fraction exceeding 0.65 AUROC.

Lang	Tier	Layer	AUROC	Drop	ρ	p
zh	high	52	.668	7.4	$-.55$	.006
fr	high	52	.672	7.0	$-.51$	.011
de	high	52	.671	7.1	$-.48$	.017
es	high	48	.661	7.4	$-.42$	.041
ja	high	48	.658	7.7	$-.37$	.075
ko	high	48	.656	7.9	$-.33$	.115
ru	high	44	.652	7.6	$-.25$	.239
pl	mid	44	.628	10.0	$-.14$	.513
ar	mid	40	.625	9.6	$-.18$	.399
id	mid	40	.621	10.0	$-.02$	.926
hi	mid	40	.618	10.3	$-.11$	.609
uk	mid	40	.614	10.7	$-.06$	.782
vi	mid	40	.607	11.4	.04	.854
th	mid	36	.594	11.8	.09	.674
bn	low	28	.553	14.1	.14	.514
te	low	28	.571	12.3	.18	.400
ta	low	28	.568	12.6	.22	.303
sw	low	28	.537	15.7	.27	.202
yo	low	24	.512	19.9	.31	.141

Table 4: Per-language zero-shot transfer on Qwen3. Layer = best transfer layer; Drop = pp below English at same layer; ρ = Spearman correlation between depth and transfer AUROC; p = associated p-value.

5.4 Transfer Method Comparison

Table 6 compares all transfer conditions across both primary models. Mean-shift produces exactly $\Delta\text{AUROC} = 0$ (as proven in §3.3). CaCLR provides negligible improvement ($+0.5$pp on Qwen3, $+0.6$pp on Llama 4; Cohen's $d = 0.04$; permutation test $p = 0.71$). Linear projection offers modest gains ($+1.8$pp on Qwen3 with 50 parallel pairs). Few-shot with 100 labels is the strongest method ($+4.4$pp on Qwen3; $d = 0.58$).

5.5 CaCLR: A Principled Negative Result

CaCLR's consistency loss ($\beta \cdot \mathcal{L}_{\text{consistency}}$) is inert. Table 7 shows the ablation: adding consistency to BCE+SB-ECE *reduces* AUROC by 0.6pp. The β sweep over three orders of magnitude (0.001–1.0) produces $<$0.4pp variation (Table 8), ruling out a tuning failure.

The mechanistic explanation: CKA analysis (§5.8) shows that cross-lingual misalignment at deeper layers is *geometric* (the correctness direction rotates across language subspaces), not merely distributional. MMD-based alignment cannot correct rotation, which is why linear projection (which learns the rotation) succeeds where CaCLR (which matches distributions) fails.

CaCLR also fails on Gemma 4 Dense (mean AUROC 0.609, -0.9pp vs. linear projection), ruling out the hypothesis that MoE routing specifically caused the failure. On Belebele and MMLU-ProX,

Model	Type	EN	ZS	ρ	Sign
Qwen3	MoE	.776	.618	$-.38$	.63
Llama4	MoE	.749	.588	$-.18$	.53
Gemma4	Dense	.761	.605	$-.44$	.68

Table 5: Three-architecture comparison. EN = best English AUROC; ZS = 19-lang mean zero-shot transfer; ρ = global Spearman; Sign = sign consistency of per-language ρ.

Method	Q3	L4	d	Labels
Zero-shot	.618	.588	—	0
Mean-shift	.618	.588	0.00	0
CaCLR	.623	.594	0.04	0 + parallel
LinProj (50)	.636	.608	0.24	$\sim$50 parallel
Few-shot 25	.644	.617	0.35	25 target
Few-shot 100	.662	.641	0.58	100 target

Table 6: Transfer method comparison (19-language mean AUROC). Q3 = Qwen3; L4 = Llama 4; d = Cohen's d vs. zero-shot.

CaCLR underperforms linear projection by -0.8pp and -1.1pp respectively, confirming the negative result generalizes across benchmarks.

5.6 Difficulty Decomposition: Inconclusive

The IRT difficulty probe achieves $R^2 = 0.52$ in English (Qwen3) but transfers with degraded quality (9-language mean $R^2 = 0.38$). Projecting out the difficulty direction changes transfer AUROC by -0.5pp on average—barely above the random-direction control (-0.3pp).

Per-language analysis shows no systematic pattern: 3/10 languages improve slightly ($+0.4$ to $+1.3$pp), 5/10 worsen (-0.3 to -2.1pp), 2/10 unchanged. The difficulty-shared component carries useful cross-lingual signal (questions hard in English tend to be hard everywhere); removing it strips away a genuine invariance. The cosine between difficulty and correctness directions (0.36 ± 0.11) confirms moderate entanglement, preventing clean separation.

5.7 Cross-Lingual Orthogonality

The correctness–confidence orthogonality reported by Miao & Ungar Miao and Ungar [2026] for English persists cross-lingually (overall mean cosine $= 0.074 \pm 0.036$; 73% of languages below 0.10). However, a resource-tier gradient emerges: high-resource mean 0.052, mid-resource 0.073, low-resource 0.108 (Spearman $\rho = -0.61$ with resource level). Yoruba on Llama 4 reaches 0.172, an outlier weakening the universality claim.

This confirms that output-based calibration faces a structural ceiling regardless of language, but this ceiling is noisier for low-resource languages where the model has less capacity to maintain the confidence–correctness dissociation.

5.8 CKA Mechanistic Analysis

CKA between English and target-language activations declines with depth (shallow mean 0.89, deep mean 0.54), confirming that deeper layers are less cross-lingually aligned. However, the correlation between CKA and transfer AUROC is weak: Pearson $r = 0.38$ on Qwen3, $r = 0.24$ on Llama 4, $r = 0.32$ combined.

A threshold analysis at CKA $= 0.35$ partially replicates: below this threshold, 61% of language-layer pairs show transfer collapse (AUROC < 0.55), vs. 22% above. But 39% of below-threshold pairs do not

Variant	AUROC	Δpp
Full CaCLR	.621	0.0
BCE + SB-ECE	.627	+0.6
BCE + Consistency	.611	−1.0
BCE only (≈ zero-shot)	.618	−0.3
Linear projection (ref.)	.636	+1.5

Table 7: CaCLR ablation (Qwen3, 19-language mean). Removing consistency *improves* performance.

β	AUROC	Δpp	β	AUROC	Δpp
0.0	.627	0.0	0.05	.623	−0.4
0.001	.626	−0.1	0.1	.621	−0.6
0.01	.625	−0.2	0.5	.618	−0.9

Table 8: CaCLR β sensitivity (Qwen3, $\alpha = 0.3$ fixed, 19-lang mean). Increasing β monotonically hurts.

collapse, and the threshold is architecture-dependent. CKA provides a suggestive but not convincing mechanistic account.

The key qualitative insight survives: cross-lingual misalignment is geometric (correctness direction rotates), explaining why MMD-based CaCLR fails while linear projection succeeds. But the quantitative relationship is too noisy for reliable prediction.

5.9 Linear Projection Data Requirements

Linear projection improves with more parallel pairs but saturates around 100 (Table 12). Critically, it fails for low-resource languages even at 200 pairs: Bengali +0.8pp, Swahili +0.6pp, Yoruba +0.6pp—within noise.

High-resource languages benefit from as few as 20 pairs (+3.1pp at 50 pairs), while low-resource languages show minimal improvement regardless of parallel data quantity. The fundamental bottleneck is not the number of parallel pairs but the quality of cross-lingual activation alignment.

5.10 Control Tasks: Capacity Artifacts

Control tasks (probes trained on randomly permuted labels) reveal that probe AUROC partially reflects capacity rather than genuine correctness signal. Table 13 shows selectivity by resource tier.

For low-resource languages, 28% of language-layer pairs have selectivity below 8pp. At the English-optimal layer, Swahili scores 0.508 (selectivity -4.4pp, *below* the random-label control) and Yoruba scores 0.487 (-6.4pp below control). These results are indistinguishable from probe memorization of random structure in high-dimensional activation spaces.

The selectivity depth gradient is positive (deeper layers have higher selectivity: 3.3pp at 0–25% depth, 9.5pp at 50–75%), reflecting that the correctness signal strengthens with depth while control AUROC remains constant (∼0.55). This means shallow-layer transfer results for low-resource languages should be interpreted with particular caution.

5.11 INT8 Quantization Validation

INT8 quantization introduces minimal distortion: mean ΔAUROC $= -0.3$pp (max -0.7pp at layer 40), activation cosine similarity $= 0.994$, layer ranking correlation $= 0.98$. Quantization is not a confound.

5.12 Cross-Benchmark Generalization

Findings partially generalize across benchmarks. On Belebele (natively multilingual reading comprehension), zero-shot transfer drops an additional 2.4pp vs. GlobalMMLU, with language rankings largely

Condition	Q3	Δ	L4	Δ
Full activations	.618	0.0	.588	0.0
Residual (diff. out)	.613	−0.5	.583	−0.5
Difficulty only	.569	−4.9	.548	−4.0
Random dir. control	.615	−0.3	.585	−0.3

Table 9: Decomposition ablation (9-language mean AUROC). Difficulty removal marginally hurts, barely above random control.

Tier	n	Qwen3	Llama4	Mean
High	8	.048	.056	.052
Mid	7	.066	.075	.073
Low	5	.100	.122	.108
All	20	.069	.079	.074

Table 10: Correctness–confidence cosine similarity by resource tier. Lower = more orthogonal. Orthogonality persists but weakens for low-resource languages.

preserved ($\rho = 0.83$). On MMLU-ProX (machine-translated MMLU), the drop is 3.9pp with lower ranking stability ($\rho = 0.69$), reflecting translation quality variation. CaCLR underperforms linear projection on both benchmarks (-0.8pp and -1.1pp respectively).

6 Analysis

Why does transfer fail for low-resource languages? Three factors compound: (1) the model's internal representations for these languages are less structured, reducing the quality of the correctness signal at any depth; (2) the cross-lingual activation alignment (measured by CKA) degrades more steeply with depth for low-resource languages (CKA at 75% depth: sw $= 0.33$, yo $= 0.28$ vs. fr $= 0.71$, zh $= 0.68$); (3) control tasks show that even the AUROC metric itself is unreliable in these settings (selectivity < 8pp).

Why does CaCLR fail? CaCLR's MMD-based consistency loss attempts to align activation *distributions* across languages. But the cross-lingual misalignment that matters for correctness probes is *geometric*: the correctness direction rotates across language subspaces at deeper layers. Distribution matching cannot correct rotation—it collapses both signal and noise. Linear projection directly estimates the rotation and therefore succeeds where CaCLR fails.

Practical guidance. For high-resource languages (en, zh, fr, de, es, ja, ko, ru): zero-shot transfer is viable with 7–8pp performance loss. Linear projection with ∼50 parallel pairs recovers an additional 1–3pp. For mid-resource languages: few-shot with 25 target-language labels provides the best cost-performance tradeoff. For low-resource languages: the zero-label dream is closed. Few-shot with 100 labels is the minimum viable approach, and even then, results should be validated against control tasks to ensure genuine correctness signal.

7 Conclusion

We present the first systematic evaluation of cross-lingual correctness probe transfer, finding that the probe-based calibration research line does not generalize beyond a small set of high-resource languages without fundamental methodological advances. Mean-shift is provably inert under standard pipelines. The depth–transferability Pareto frontier is architecture-dependent, precluding universal layer selection. CaCLR—the natural distributional-matching approach—fails because cross-lingual misalignment is

Model	r	p	Sig.	Total
Qwen3	.38	.002	5/19	19
Llama4	.24	.041	2/19	19
Combined	.32	.001	7/38	38

Table 11: CKA vs. transfer AUROC correlation. Sig. = languages with individually significant correlation.

n	AUROC	Δpp	% Improved
10	.608	-1.0	26%
20	.621	$+0.3$	42%
50	.636	$+1.8$	55%
100	.643	$+2.5$	63%
200	.647	$+2.9$	68%

Table 12: Linear projection AUROC by number of parallel pairs (19-language mean). Saturates at $\sim$100 pairs.

geometric, not distributional. Few-shot adaptation with 25–100 labels is the only viable path for the majority of the world's languages.

Limitations. Our study is restricted to MCQA benchmarks. Open-ended generation may exhibit different calibration patterns. INT8 quantization, while validated, may interact differently with other probe architectures. The resource-tier taxonomy is coarse—individual language pairs may deviate based on typological proximity rather than resource level alone.

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Tier	n	Probe	Ctrl	Sel.	Min
High	7	.660	.548	17.2	10.4
Mid	7	.614	.546	13.8	8.2
Low	5	.548	.544	8.4	4.8
All	19	.618	.546	13.8	4.8

Table 13: Control task selectivity (pp) by resource tier. Probe = real-label AUROC; Ctrl = random-label AUROC; Sel. = selectivity (Probe $-$ Ctrl); Min = minimum selectivity in tier.

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